

From Deterministic Chaos to Quantum Potential: A Critical Evaluation of Fisher Information Emergence in Invariant Set Theory

This report conducts a deep, adversarial investigation into whether a deterministic microscopic dynamics, specifically within the Invariant Set Theory (IST) framework, can rigorously generate the osmotic velocity field and its associated Fisher information structure that underpin Bohmian quantum mechanics. The inquiry is guided by the core question: Can a deterministic, fractal-based invariant set evolve in such a way that its coarse-grained description yields the exact hydrodynamic form $u(x,t) = (\hbar/2m)\nabla \ln \rho(x,t)$ or its equivalent variational principle, $(\hbar^2/8m)IF[\rho]$, without inserting quantum mechanical assumptions by construction? This analysis critically examines candidate mechanisms, evaluates existing mathematical results, clarifies key constraints, and assesses the overall viability of this foundational pathway for IST. The research adheres to strict determinism, avoids the introduction of fundamental stochasticity, and demands mathematical rigor in any proposed coarse-graining or continuum limit.

The Stochastic Mechanics Blueprint and Its Deterministic Limitations

Nelson's stochastic mechanics provides the most direct and mathematically detailed blueprint for deriving the quantum potential from a more fundamental microscopic process ¹⁴⁴¹⁴⁵. Understanding its precise assumptions and limitations is crucial for evaluating any attempt to replace its stochastic foundation with a deterministic one. The theory aims to derive the Schrödinger equation from principles of classical physics augmented by the assumption of an underlying random process ¹⁴⁴⁴⁸¹. The central concept is that a particle's trajectory is not a smooth curve but undergoes a continuous, non-differentiable random motion, described by a stochastic differential equation (SDE)

The derivation begins by considering two velocities defined along a particle's random path: the forward drift velocity, $v^+(x,t)=\lim_{dt\rightarrow 0^+}\langle dx/dt\rangle$, and the backward drift velocity, $v^-(x,t)=\lim_{dt\rightarrow 0^+}\langle dx/(-dt)\rangle$. These define the current velocity, $v=(v^++v^-)/2$, and the osmotic velocity, $u=(v^+-v^-)/2$ ¹⁵². The osmotic velocity is the source of all distinctly quantum effects in this framework ¹⁵². The probability density $\rho(x,t)$ of finding the particle at position x at time t evolves according to the continuity equation, $\partial_t\rho=-\nabla\cdot(\rho v)$. By applying Ito's lemma to the SDE, one can derive a pair of coupled partial differential equations for the forward and backward velocities. The final step involves imposing a condition of detailed balance, which relates the probability densities forward and backward in time and is analogous to energy conservation. This condition allows the osmotic velocity to be expressed as $u=(\hbar/2m)\nabla\ln\rho$, where the constant $\hbar$ emerges as the diffusion coefficient, specifically $D=\hbar/(2m)$ ¹⁴⁴¹⁴⁵.

The table below outlines the key assumptions and their role in Nelson's derivation.

Assumption	Role in Derivation	Status for Deterministic IST
Stochastic Differential Equation (SDE)	Defines particle trajectories as non-differentiable paths, providing the mathematical foundation for forward/backward velocities ¹⁴⁴ .	Fundamentally Incompatible. IST requires deterministic trajectories.
Diffusion Coefficient ($D=\hbar/2m$)	Directly sets the magnitude of the osmotic velocity. It is identified with the ratio of Planck's constant to twice the particle mass ¹⁴⁵ .	Critical Challenge. Must emerge from geometry, not be inserted as a parameter.
Time-Reversal Asymmetry	Inherent in the SDE, leading to distinct forward and backward drifts, which is necessary for the osmotic term ¹⁴⁵ .	Plausible. A deterministic system could have time-asymmetric components, though the mechanism differs.
Ito Calculus	Provides the correct calculus for handling the non-differentiable paths and deriving the evolution equations ¹⁴⁵ .	Mathematical Tool. A different calculus might be needed for deterministic fractals.
Detailed Balance Condition	A physical postulate relating the forward and backward probabilities, effectively enforcing energy conservation ¹⁴⁴ .	Conceptually Plausible. Could correspond to a constraint on the invariant set's geometry.

The primary obstacle for IST is the reliance on stochasticity. Simply replacing the noise term in Nelson's SDE with a deterministic chaotic forcing function is insufficient. The challenge is not merely to make deterministic chaos "look" stochastic in some statistical sense, but to reproduce the specific algebraic structure of the osmotic velocity term, $u=(\hbar/2m)\nabla\ln\rho$. While deterministic systems can exhibit statistical behavior similar to diffusion, the critical question is whether they can generate this particular functional form of the velocity field. Some work has explored extending Nelson's mechanics to fractal time frameworks using fractal calculus, suggesting a conceptual overlap between the domains ⁴⁹². However, this remains speculative and does not provide a proof for a fully deterministic system.

Furthermore, attempts to reconstruct quantum mechanics from stochastic foundations have faced challenges. Some formulations have been shown to be inequivalent to the standard Schrödinger equation ¹⁴⁷, and certain predictions may differ empirically ¹⁴⁸. This suggests that the connection between stochastic processes and quantum mechanics, while profound, is subtle and not guaranteed to hold under arbitrary modifications, such as replacing fundamental randomness with deterministic chaos. Therefore, while Nelson's theory provides a valuable target—identifying the precise mathematical object to be replicated—it also highlights the immense difficulty of achieving this goal within a purely deterministic framework. Any successful derivation in IST would need to find a novel mechanism that produces the same mathematical outcome as Nelson's detailed balance condition, but through the geometric and topological properties of its fractal invariant set.

Information Geometry as a Uniqueness Constraint for Quantum Emergence

An alternative route to the quantum potential bypasses microscopic dynamics altogether, instead appealing to principles of information geometry and entropic inference. Pioneered by Caticha, Hall, and Reginatto, this approach derives the laws of quantum mechanics by applying the method of maximum entropy to a classical ensemble of particles subject to certain constraints ^{5 61 226}. This perspective reframes the problem from "how can a dynamical system produce Q?" to "what is the unique geometric structure that a consistent theory of inference must possess?".

The core idea is to construct an action principle based on information-theoretic concepts. The resulting dynamics are driven by the principle of maximum ignorance (entropy), subject to constraints representing physical conditions like energy conservation ³³⁷. A key insight is that the Fisher information plays a central role. The Fisher information, $IF[\rho] = \int \rho |\nabla \ln \rho|^2 dx$, quantifies the amount of information that an observable random variable X carries about an unknown parameter θ upon which its probability distribution depends ¹⁰³¹³⁸. In the context of a particle ensemble, it measures the "sharpness" of the probability distribution $\rho(x,t)$ ¹⁴². Reginatto demonstrated that by adding a term proportional to the Fisher information to a classical action, one can derive the Schrödinger equation ^{12 13 323}. Specifically, minimizing the Fisher information, or equivalently maximizing the smoothness of the ensemble's evolution, leads directly to the form of the quantum potential ²⁴¹²⁷⁸.

What makes this approach particularly powerful is the remarkable uniqueness of the Fisher information metric. It is not an arbitrary choice. Cencov's theorem states that the Fisher-Rao metric is the unique Riemannian metric that can be defined on a space of probability distributions (a statistical manifold) that is invariant under mappings between experiments that preserve all statistical information ⁴³³⁴⁴⁰. This geometric characterization means that any theory that seeks to describe the evolution of probabilities in a manner consistent with information processing must, in some sense, use this specific metric ⁵³⁴⁵³⁵. For IST, this has a profound implication: if the theory is to successfully generate the quantum potential, its emergent coarse-grained geometry must, in a very precise way, correspond to the Fisher-Rao geometry. The task is not to show that deterministic chaos can produce *some* functional that looks like Fisher information; it is to show that the geometry of the invariant set naturally gives rise to a structure that satisfies the axioms of information geometry, thereby forcing the Fisher-Rao metric as the only consistent choice.

This information-geometric perspective shifts the burden of proof onto the IST framework. Instead of searching for a dynamical process that generates an indefinite stress tensor, the focus becomes on identifying a natural information-theoretic quantity on the fractal invariant set. The p-adic integers, which are homeomorphic to a Cantor set, offer a promising arena for exploring this ²⁶⁴⁴⁶. One could investigate whether a suitable conditional probability measure on the branches of the p-adic Cantor set leads to a discrete Fisher information that, in a continuum limit, converges to the standard continuum form. The challenge remains formidable, as the entire apparatus of information geometry is built upon the language of smooth manifolds and probability densities, concepts that are ill-defined on a measure-zero fractal set. Bridging this gap would require developing a novel form of information geometry adapted to non-smooth spaces, potentially drawing on concepts from Dirichlet forms or Γ -calculus on fractals ¹⁸¹²⁷⁹. Nevertheless, this path offers a more constrained and potentially more elegant route than pure dynamical homogenization, as it leverages the powerful uniqueness properties of information geometry to guide the search for a deterministic origin of quantum structure.

Deterministic Homogenization and Fractal Coarse-Graining

The most direct strategy for replacing the stochastic machinery of Nelson's mechanics with a deterministic one lies in the field of deterministic homogenization and coarse-graining. This area of mathematics investigates how the collective behavior of a complex, deterministic system can converge to a simpler, often stochastic, effective description in a scaling limit ^{76 355}. The central question is whether such techniques can be used to generate the specific osmotic velocity field, $u=(\hbar/2m)\nabla\ln\rho$, rather than just generic diffusive behavior.

Recent progress has shown that deterministic fast-slow systems, where a slow variable of interest evolves under the influence of a rapidly fluctuating fast variable, can indeed have their slow dynamics converge weakly to a stochastic diffusion process described by an Itô SDE ⁴¹²⁴¹³⁵⁶¹. The drift and diffusion coefficients of the limiting SDE can be calculated explicitly from averages over the fast dynamics ⁴⁰⁸. This demonstrates that deterministic chaos can, in a well-defined mathematical sense, look stochastic. Furthermore, anomalous diffusion has been shown to arise from advection by a deterministic vector field with a fractal structure, achieved by repeatedly inserting shear flows at infinitely many scales ⁴¹⁰⁴¹¹. These results establish that deterministic systems can produce effective transport phenomena that share features with the underlying process in Nelson's mechanics.

However, a significant gap exists between producing a generic diffusion limit and generating the *specific* structure of the osmotic term. The diffusion coefficient D in the homogenized SDE arises from correlation functions of the fast chaotic dynamics ⁴⁰⁸. In IST, this would correspond to correlations within the fractal invariant set. The critical challenge is twofold: first, to show that this emergent diffusion coefficient is precisely $(\hbar/2m)$, linking the microscopic dynamics to a universal constant; and second, to ensure that the resulting stress tensor for the coarse-grained fluid flow has the indefinite signature characteristic of the quantum potential, rather than the positive semidefinite signature of the classical Reynolds stress tensor ³⁰¹. Ordinary variance tensors are inherently positive semidefinite, whereas the quantum stress tensor, which governs the quantum potential, is not. A naive coarse-graining of a deterministic system would likely produce a positive-definite contribution, failing to capture the essence of quantum interference.

The Gemini Flash argument correctly identifies this failure of the "positive variance" route, elevating the importance of the osmotic/informational route. It implies that any viable deterministic derivation must go beyond simple variance calculations and produce a term that is fundamentally tied to the logarithmic gradient of the density, $\nabla \ln \rho$. This points towards mechanisms involving forward and backward decomposition of trajectories, a concept explored in the context of stochastic mechanics and symbolic dynamics ¹⁵²³⁵⁶. If a deterministic system possesses a natural pairing of past and future states or a branching structure in its phase space, one could define a difference between forward and backward evolving measures, which might then yield an osmotic-like velocity in a coarse-grained limit. The development of quantitative homogenization techniques for interacting particle systems suggests that rigorous limits for complex deterministic models are becoming tractable ²⁰⁴⁵²⁴.

Ultimately, proving that such a limit yields the precise Fisher information action, $(\hbar^2/8m)I_F[\rho]$, remains an open problem. It would require a new theorem establishing a connection between the geometric properties of the invariant set (e.g., its Hausdorff dimension, self-similarity ratios) and the emergence of this specific information-geometric functional. The success of this program hinges on moving beyond the general statement that "chaos can look stochastic" to a more specific result demonstrating that a particular class of deterministic fractal systems generates the unique geometric structure demanded by information geometry.

Invariant Set Theory-Specific Structures as Potential Mechanisms

While general theories of stochastic mechanics and homogenization provide useful analogies, the ultimate test for the IST program is whether its own unique mathematical structures can serve as the foundation for generating the required quantum geometry. Palmer's framework posits that the universe's state space contains a non-computable, fractal invariant set, I , of measure zero, on which all physically realizable states lie ¹⁷⁰³⁸⁴. Several features of this set, and the mathematics developed to describe it, represent concrete candidates for the sought-after emergence mechanism.

First, **p-adic geometry** has been explicitly proposed as a natural language for the invariant set's topology ²⁶⁵⁰⁹. The ring of p-adic integers, $\mathbb{Z}_p$, is topologically equivalent to a Cantor set ⁴⁴⁶. This provides a rigorous bridge between the abstract fractal geometry

of IST and a well-established branch of number theory and analysis ⁵². P-adic analysis possesses its own calculus, integration theory (based on Haar measure rather than σ -algebras), and spectral theory ^{54 488}. Research has established the existence of Laplacian operators and diffusion processes on p-adic manifolds ⁴⁶⁰⁵⁴⁶⁵⁴⁷. This opens a concrete pathway: one could define a suitable action or energy functional on a p-adic configuration space and investigate its continuum limit. If this limit reproduces the Fisher information structure, it would provide strong support for the IST program. The challenge lies in connecting the abstract p-adic structure to the familiar real numbers of our macroscopic world, a process sometimes referred to as p-adicization ³⁰⁶.

Second, **symbolic dynamics** offers a way to describe the evolution on the invariant set ³⁵⁶³⁶⁰. Since the invariant set is a Cantor set of state-space trajectories, its structure can be represented by infinite sequences of symbols, where each symbol corresponds to a choice of branch at a given level of refinement ³⁵⁷. This tree-like structure suggests a mechanism for generating an osmotic velocity. One could consider the probability flow through this tree. The difference between the probability of a path evolving forward versus backward through a branching point could, in a coarse-grained description, manifest as a velocity field proportional to $\nabla \ln \rho$. This connects directly to the forward/backward decomposition idea. However, this requires developing a robust measure-theoretic formalism on the Cantor set, a non-trivial task ³⁴⁶. The existence of invariant measures for quantum trajectories provides some precedent for defining such measures in related contexts ³⁰⁸³⁶⁷.

Third, the study of **Dirichlet forms on fractals** presents another promising avenue ¹⁸¹. Dirichlet forms provide a powerful analytical tool that links stochastic processes (diffusions) to quadratic energy functionals, which are the basis for Laplacians and heat kernels ⁴⁸⁶. The classical Dirichlet integral, $\int |\nabla f|^2 dx$, is closely related to the Fisher information functional. Research has explored the construction of Dirichlet forms and associated Laplacians on various fractals, including those with p-adic structure ⁵⁶³⁵⁶⁷⁵⁶⁹. If a suitable Dirichlet form can be defined on the IST invariant set, its associated energy functional might be the desired Fisher information action. The work on fractal-fractional calculus, which combines fractal geometry with fractional operators, further suggests that the mathematical tools for analyzing dynamics on such sets are being actively developed ⁴⁹⁰. A major hurdle is that many standard analytical tools, like the Lebesgue integral, do not generalize straightforwardly to the non-Archimedean setting of p-adic analysis, requiring the development of new mathematical machinery ⁴⁸⁸.

In summary, IST is not devoid of mathematical structure. It possesses rich connections to p-adic geometry, symbolic dynamics, and fractal analysis. However, none of these structures are currently developed in the published literature to the point where they can be shown to generate the specific Fisher information structure of Bohmian mechanics. They represent fertile ground for new research but are not yet a completed mechanism. The IST program would need to synthesize these disparate areas of mathematics into a coherent whole capable of producing the required result.

Re-evaluating Key Constraints: Hudson's Theorem and the Origin of $\hbar$

Two critical constraints shape the feasibility of the IST program: the nature of obstructions like Hudson's theorem and the mysterious origin of Planck's constant, $\hbar$. A careful re-evaluation of these elements is necessary to distinguish genuine mathematical barriers from misconceptions or profound theoretical puzzles.

The initial adversarial review pointed to Hudson's theorem as a significant obstruction to the "naive Route B," which attempted to equate the quantum stress tensor with a classical positive variance (Reynolds stress). Hudson's theorem, however, requires precise interpretation. It states that for a continuous-variable quantum system, the Wigner function of a pure state is everywhere non-negative if and only if the state is a Gaussian wavepacket ¹⁵⁵¹⁵⁶²⁸⁰²⁸³. The Wigner function is a quasiprobability distribution in *phase space*, not a classical probability density in configuration space ²⁸¹. The osmotic velocity, $u = (\hbar/2m)\nabla \ln \rho$, is a concept from the hydrodynamic formulation of quantum mechanics, which is based on the configuration-space probability density $\rho(x,t)$ ⁴. Therefore, Hudson's theorem does not directly forbid the existence of a classical ensemble whose coarse-grained hydrodynamics matches the Bohmian form. The theorem constrains the full phase-space representation of quantum states, while the osmotic velocity is a consequence of the configuration-space density alone. The true obstruction highlighted by the Gemini Flash argument is the fact that the quantum stress tensor is indefinite, whereas the classical momentum flux (related to the square of velocity fluctuations) is positive semidefinite ¹⁵³. This makes the "osmotic/Fisher route" necessary, but Hudson's theorem itself is not the direct barrier. It serves to strengthen the case for seeking a more subtle, information-geometric origin for the quantum potential rather than a simple statistical one.

The second, and perhaps greater, challenge is the origin of the constant $\hbar$. In both Nelson's stochastic mechanics and the information-theoretic derivations, $\hbar$ appears as a parameter: either as the diffusion constant or as a Lagrange multiplier enforcing a constraint [12, 144]. In a deterministic, ontological theory like IST, such a constant cannot be introduced arbitrarily. It must emerge from the intrinsic geometric or dynamic properties of the invariant set. The report's user is explicit that dimensional analysis alone is insufficient to derive $\hbar$ [22, 71]. This problem is deeply connected to the quest for a theory of quantum gravity, where understanding the origin of $\hbar$ is a central theme [22, 28].

For IST, this means that the scale of the Planck length, or some other fundamental geometric property of the fractal invariant set, must be related to the emergence of $\hbar$. One might hypothesize that the self-similarity ratio of the Cantor set, or a curvature scale associated with its embedding, when properly normalized, produces a quantity with the dimensions of action. This would likely require extending IST with a specific scaling hypothesis or discovering a unique normalization condition inherent to its geometric structure. Without such a mechanism, any model would remain incomplete, producing a velocity field of the correct form but with an arbitrary prefactor c instead of the specific $(\hbar/2m)$. This issue is arguably the most difficult remaining puzzle for the IST program and represents a major obstacle to its completion. Until a plausible mechanism for generating $\hbar$ from first principles is found, the theory cannot be considered fully realized.

Synthesis and Path Forward for the IST Program

After a comprehensive adversarial examination of the available mathematical frameworks and IST-specific structures, the status of the hypothesis—that a deterministic invariant-set dynamical system can generate the osmotic/Fisher-information structure required for Bohmian mechanics—is best summarized as follows:

Bottom-line Verdict: B. PLAUSIBLE BUT REQUIRES A NEW THEOREM.

There is no existing mathematical theorem that establishes this emergence, ruling out option (A). However, the combination of insights from information geometry, deterministic homogenization, and IST's unique mathematical structures provides a coherent and compelling, albeit highly non-trivial, constructive path forward, making option (B) the most accurate assessment. Option (C), requiring genuinely new mathematical structures, is possible but seems less likely than the prospect of building

upon existing advanced mathematics. A decisive no-go result (D) has not been established, meaning the possibility remains open, though unproven.

The literature proves several relevant facts:

- Nelson's stochastic mechanics provides a rigorous derivation of the quantum potential starting from a diffusion process, identifying the osmotic velocity as $u = (\hbar/2m)\nabla \ln \rho$ ¹⁴⁵¹⁵².
- Recent advances in deterministic homogenization show that chaotic or fast-slow deterministic systems can converge weakly to stochastic diffusion processes, demonstrating that deterministic chaos can mimic stochastic behavior ⁴⁰⁸⁵⁶¹.
- The Fisher information metric is uniquely characterized by information-theoretic axioms, such as Cencov's theorem, which identifies it as the unique Riemannian metric invariant under sufficient statistics ⁴³³⁴⁴⁰. This uniqueness is a powerful constraint on any emergent theory.
- The IST framework contains rich mathematical structures, including p-adic geometry, symbolic dynamics, and fractal Dirichlet forms, which are potential candidates for generating the required information-geometric structure ¹⁸¹³⁵⁶⁵⁰⁹.

The literature rules out several naive approaches:

- A direct substitution of stochasticity with deterministic chaos in Nelson's framework is insufficient without a deeper, more specific mechanism ¹⁴⁴.
- A naive coarse-graining of a classical ensemble, based on positive variance (Reynolds stress), fails because the resulting stress tensor is positive semidefinite, while the quantum stress tensor is indefinite ¹⁵³.
- Dimensional analysis alone is not a sufficient derivation of the constant $\hbar$ ^{22 71}.

The Gemini Flash review correctly identified the failure of the "positive variance" route and the necessity of the osmotic/informational route. However, it overstated the relevance of Hudson's theorem, which primarily constrains phase-space quasiprobabilities (Wigner functions) and does not directly obstruct the generation of the configuration-space hydrodynamic equations ¹⁵⁵²⁸⁰.

The strongest constructive route currently available is a synthesis of the information-geometric and IST-specific ideas. It would proceed as follows: 1. Formalize the IST invariant set as a p-adic Cantor set, leveraging its topological equivalence to the p-adic integers ^{26 446}. 2. Develop a conditional probability measure on the branches of this set, representing the local, deterministic dynamics. 3. Define an appropriate discrete information-theoretic quantity (analogous to Fisher information) on this measure. 4.

Prove that extremizing this discrete quantity leads to a discrete Euler-Lagrange equation governing the coarse-grained particle motion. 5. Finally, demonstrate that in a suitable continuum limit, this discrete action converges to the standard Fisher information functional, $(\hbar^2/8m)IF[\rho]$, with $\hbar$ emerging from the intrinsic scale of the p-adic geometry.

The strongest no-go argument would be a theorem proving that for any sufficiently chaotic deterministic system with a positive invariant measure, the effective coarse-grained stress tensor is necessarily positive semidefinite, thereby precluding the indefinite structure required for the quantum potential. Such a result would force the IST program to rely entirely on the more abstract information-geometric route.

The decisive missing theorem in the IST framework is one that connects the geometric properties of its fractal invariant set to the emergence of a Fisher-Rao metric in the coarse-grained configuration space. This would be the cornerstone of the program.

The following table summarizes the status of claims central to the investigation.

Claim	Status	Existing Theorem/Source	Assumptions	BM-IST Applicability	What Remains to Prove
Osmotic velocity $u = (\hbar/2m)\nabla \ln \rho$ emerges from a diffusion process.	Established	Nelson's Stochastic Mechanics 145152	Stochastic trajectories, Ito calculus, detailed balance.	Not applicable (stochastic)	None.
Deterministic fast-slow systems can have diffusive limits.	Established	Kelly et al., Melbourne & Stuart 408409561	Fast/slow separation, hyperbolic fast dynamics.	Plausible	That the <i>specific</i> osmotic term emerges, not just generic diffusion.
Fisher information is the unique Riemannian metric from info. axioms.	Established	Cencov's Theorem 433440	Invariance under sufficient statistics.	Plausible	That the geometry of the IST set satisfies these axioms.
The Wigner function of a non-Gaussian pure state is negative.	Established	Hudson's Theorem 155280	Pure state in continuous-variable system.	Relevant but not a direct obstruction.	Clarify distinction between phase-space and configuration-space hydrodynamics.
The p-adic integers are topologically a Cantor set.	Established	General Topology 446	Definition of p-adic topology and Cantor set.	High (forms basis for p-adic modeling of IST)	A complete theory of analysis on this set to produce the desired action.
IST invariant set is a fractal subset of state space.	Assumed (Postulate)	Palmer, IST Postulate 170510	Cosmological Invariant Set Hypothesis.	Core assumption of the theory.	None.

The shortest credible path to a definitive answer would be to prove a new theorem. A candidate falsification test would be to construct a concrete, explicit deterministic toy model, such as a piecewise linear map on a Cantor set, and rigorously compute its coarse-grained stress tensor. If the result is always positive semidefinite, it constitutes a strong no-go. Conversely, if the model can be made to produce an indefinite stress tensor proportional to $\nabla \ln \rho$, it would be a landmark result supporting the IST program.

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